* 0000800000001 * DFD

CAMBRIDGE International Education

Cambridge IGCSE™

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

* 4 9 0 1 9 5 4 8 8 0 *

MATHEMATICS **0580/11**

Paper 1 Non-calculator (Core) **May/June 2026**

1 hour 30 minutes

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly.

INFORMATION

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages.

DC (WW/SW) 355670/4

List of formulas

Area, A, of triangle, base b, height h.	$A = \frac{1}{2}bh$
Area, A, of circle of radius r.	$A = \pi r^2$
Circumference, C, of circle of radius r.	$C = 2\pi r$
Curved surface area, A, of cylinder of radius r, height h.	$A = 2\pi rh$
Curved surface area, A, of cone of radius r, sloping edge l.	$A = \pi rl$
Surface area, A, of sphere of radius r.	$A = 4\pi r^2$
Volume, V, of prism, cross-sectional area A, length l.	$V = Al$
Volume, V, of pyramid, base area A, height h.	$V = \frac{1}{3}Ah$
Volume, V, of cylinder of radius r, height h.	$V = \pi r^2 h$
Volume, V, of cone of radius r, height h.	$V = \frac{1}{3}\pi r^2 h$
Volume, V, of sphere of radius r.	$V = \frac{4}{3}\pi r^3$

Calculators must **not** be used in this paper.

1

1.694	$\sqrt{18}$	6	6.666	7	$\sqrt{225}$	25	27	39	56	1000

From the list of numbers, write down

(a) an even number

.... [1]

(b) a factor of 28

.... [1]

(c) a square number

.... [1]

(d) a prime number

.... [1]

(e) an irrational number.

.... [1]

2 **(a)** Bella is b years old.
Alan is 5 years older than Bella.

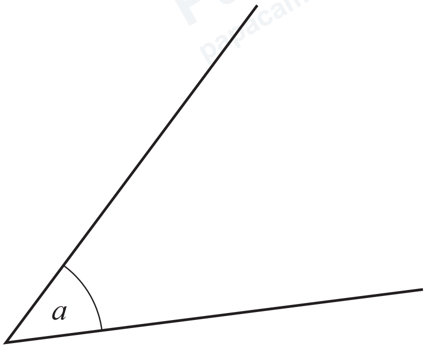
Write an expression, in terms of b, for how old Alan is.

.... [1]

(b) Write an expression for the total cost, in cents, of x bags of sweets costing 65 cents each.

.... cents [1]

3



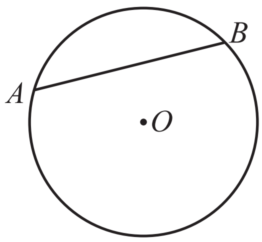
(a) Measure angle a.

.... [1]

(b) Write down the mathematical name for this type of angle.

.... [1]

4 (a) Points A and B lie on a circle, centre O, as shown in the diagram.



Write down the mathematical name for line AB.

.... [1]

(b) A quadrilateral has

- four equal sides
- no right angles
- opposite angles that are equal.

Write down the mathematical name for this quadrilateral.

.... [1]

5 **(a)** Write $\frac{1}{2}$ as a decimal.

.... [1]

(b) Write $\frac{9}{10}$ as a percentage.

.... % [1]

(c) Write 0.15 as a fraction in its simplest form.

.... [1]

6 Triangle ABC has sides $AC = 7.4\,\text{cm}$ and $BC = 4.6\,\text{cm}$.

Using a ruler and compasses only, construct triangle ABC.
Leave in your construction arcs.
The side AB has been drawn for you.

[2]

7 **(a)** Two fair 4-sided dice have faces numbered 1 to 4.
The dice are thrown, and the numbers are multiplied to give a total.
The table shows all the possible results.

dice *B*

		1	**2**	**3**	**4**
	1	1	2	3	4
dice *A*	**2**	2	4	6	8
	3	3	6	9	12
	4	4	8	12	16

Find the probability the total is

(i) 12

.... [1]

(ii) 5

.... [1]

(iii) a number that is less than 20.

.... [1]

(b) A biased 6-sided dice, numbered 1 to 6, is thrown.
The probability of scoring the number 6 is 0.2 .

Work out the probability of scoring a number less than 6.

.... [1]

8 In a town, the lowest temperature one night is $-12.6\,°\text{C}$.
The highest temperature the next day is $25.7\,°\text{C}$.

Calculate the difference between these two temperatures.

.... °C [1]

9 When Ahmed gets out of bed he takes

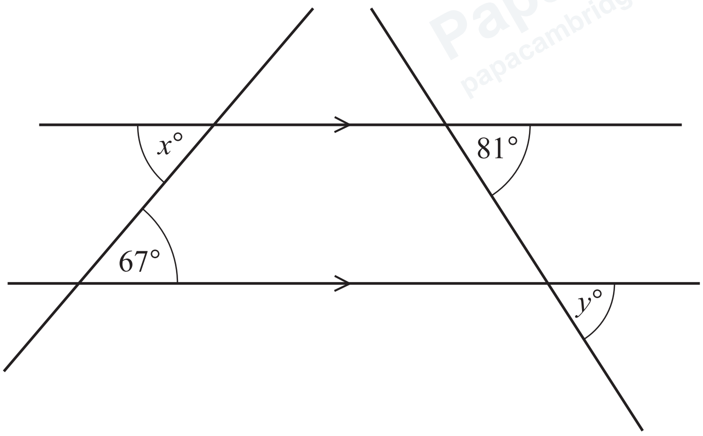
- 12 minutes to get dressed
- 26 minutes to eat breakfast
- 37 minutes to walk to the station.

Ahmed must be at the station by 09 45.

Work out the **latest** time Ahmed can get out of bed.

.... [2]

10 The diagram shows two straight lines intersecting two parallel lines.



NOT TO SCALE

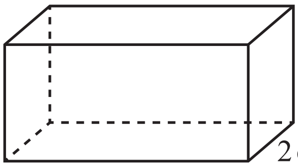
Complete each of these statements with the correct geometrical reason.

$x = 67$ because ....

$y = 81$ because ....

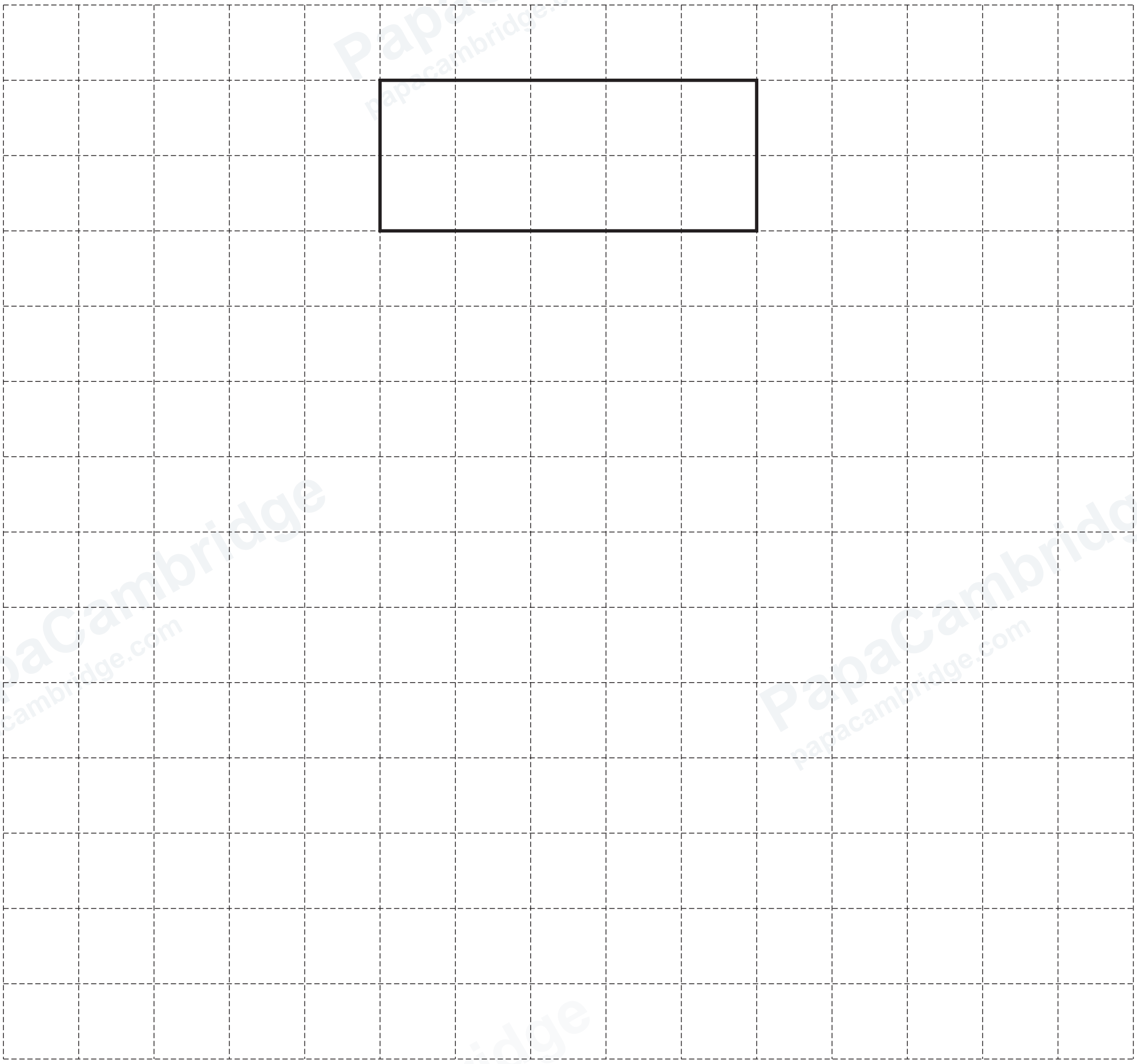
[2]

11



NOT TO SCALE

On the $1\,cm^2$ grid, complete a net of this cuboid.
One face has been drawn for you.



[3]

12 The stem-and-leaf diagram shows the lengths, in centimetres, of 32 wooden planks.

23	3 4 4 5 6 6 8 9
24	0 0 1 1 1 1 3 6 8 8 9
25	2 3 5 6 7 9 9 9
26	0 0 1 2 4

key : 23 | 3 represents 233 cm

(a) Find the range.

.... cm [1]

(b) Find the mode.

.... cm [1]

(c) Find the median.

.... cm [1]

(d) Find the fraction of all these planks which are less than 236 cm.
Write your answer as a fraction in its simplest form.

.... [2]

13 By writing each number in the calculation correct to 1 significant figure, work out an estimate for the value of

$$\frac{9.57 \times 3.84}{6.72 + 1.13}.$$

.... [2]

14 A bag contains 45 balls.
The balls are red or green.
The balls are in the ratio red balls : green balls = 2 : 3.

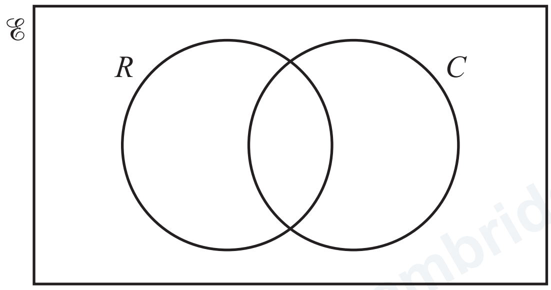
Carlos puts 4 more red balls and x more green balls into the bag.
The balls are now in the ratio red balls : green balls = 1 : 2.

Find the value of x.

x = ... [4]

15 **(a)** 60 people complete a survey about hobbies.

41 people like reading books (R).
31 people like cookery (C).
11 do not like reading books and do not like cookery.



(i) Complete the Venn diagram.

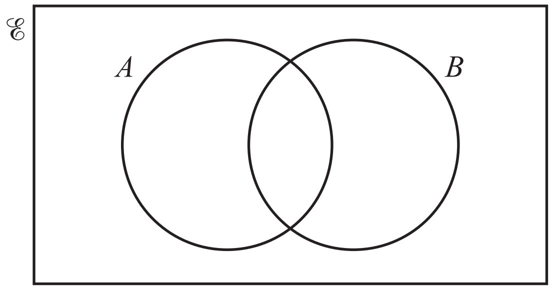
[2]

(ii) One of these 60 people is picked at random.

Find the probability that they like cookery but they do not like reading books.

.... [1]

(b)



Shade $A \cap B$. [1]

16 Buses at a station go to Alve or to Riba.
Buses leave every 16 minutes for Alve.
Buses leave every 28 minutes for Riba.
At 08 25 a bus for Alve and a bus for Riba leave the station together.

Find the next time a bus for Alve and a bus for Riba leave the station together.

.... [3]

17 Work out.

$$2\frac{1}{6} - 1\frac{3}{5}$$

Write your answer as a fraction in its simplest form.

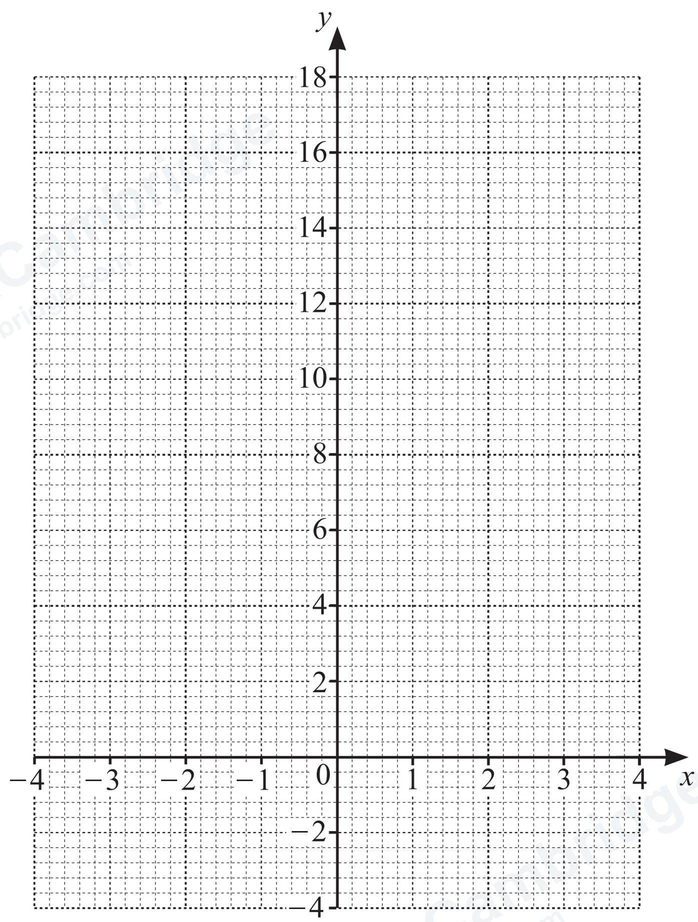
.... [3]

18 **(a)** Complete the table of values for $y = x^2 + x - 3$.

x	–4	–3	–2	–1	0	1	2	3	4
y	9		–1	–3	–3	–1		9	

[2]

(b) On the grid, draw the graph of $y = x^2 + x - 3$ for $-4 \leqslant x \leqslant 4$.



[4]

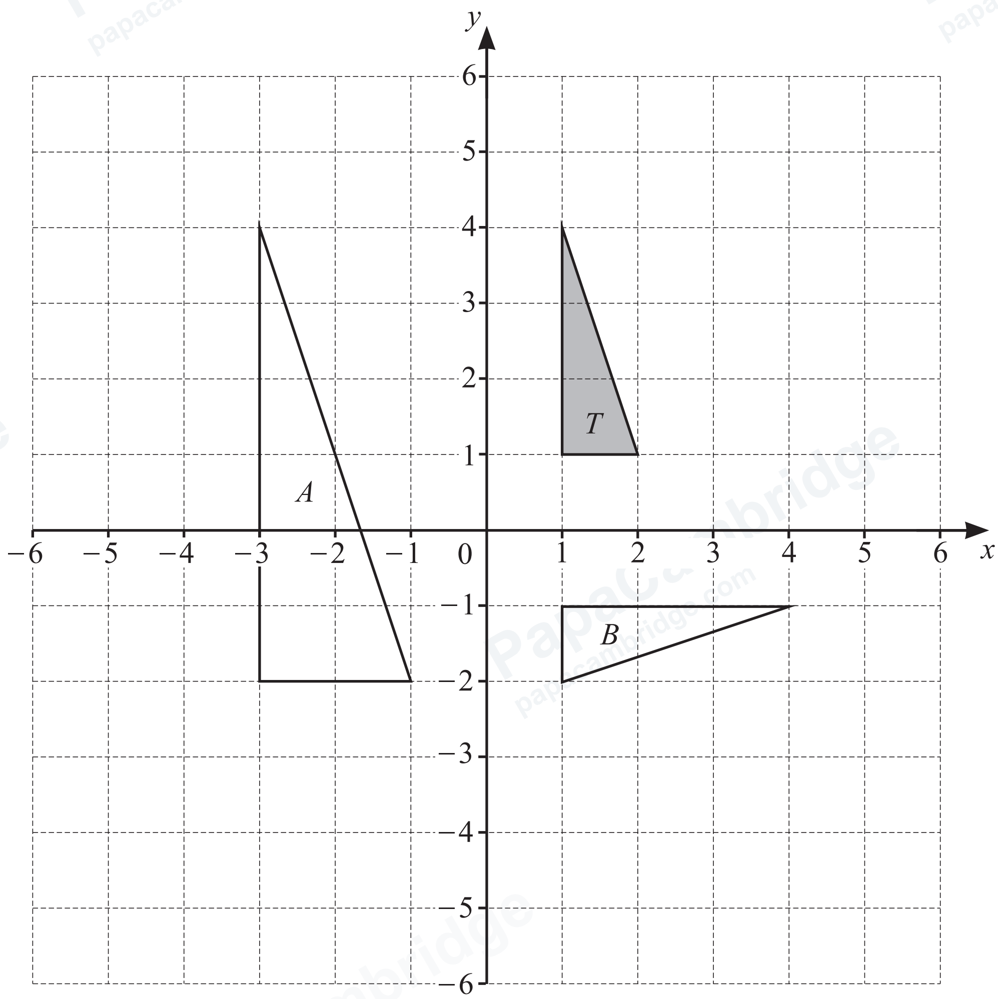
(c) Write down the equation of the line of symmetry of the graph.

.... [1]

(d) Use your graph to solve the equation $x^2 + x - 3 = 0$.

$x =$ .... or $x =$ .... [2]

19



(a) On the grid, draw the image of triangle T after a translation by the vector $\begin{pmatrix} 3 \\ -6 \end{pmatrix}$. [2]

(b) On the grid, draw the image of triangle T after a reflection in the line $y = -1$. [2]

(c) Describe fully the **single** transformation that maps triangle T onto triangle A.

....

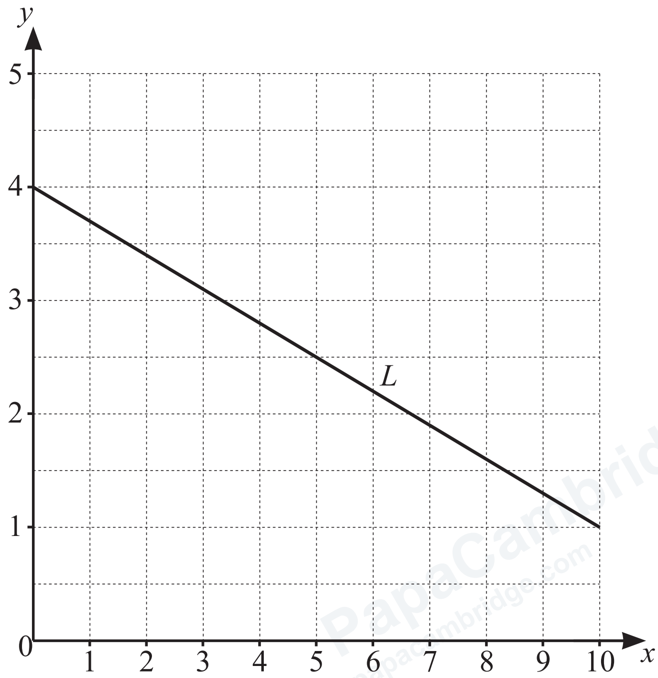
.... [3]

(d) Describe fully the **single** transformation that maps triangle T onto triangle B.

....

.... [3]

20 The diagram shows line L on a grid.



Find the equation of line L in the form $y = mx + c$.

$y =$ .... [3]

21 **(a)** Make T the subject of the formula.

$P = Ty$

$T =$ .... [1]

(b) Expand the brackets and simplify.

$(x+8)(x-5)$

.... [2]

22 The time, t seconds, for the winner of a race is 12.7 seconds, correct to 3 significant figures.

Complete this statement about the value of t.

.... $\leqslant t <$ .... [2]

23 Solve the simultaneous equations.

$$3x + 4y = 1$$
$$2x - 3y = 12$$

$x =$ ....

$y =$ ....
[4]